

Emmanuel Suárez Acevedo

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Work Experience

Graduate Researcher

Cornell University

New York, NY

August 2024 – Present

- * Developing formal verification techniques for randomized concurrent systems with fair scheduling guarantees, targeting correctness of distributed consensus protocols
- * Designed programming language for automated verification of quantitative network behavior (e.g., *is all traffic delivered within 9ms?*) and presented paper at PLDI 2026

Graduate Researcher

University of Pennsylvania

Philadelphia, PA

August 2022 – April 2024

- * Developed type system for statically tracking computational effects and resource demands, proving dead code elimination preserves program behavior (OOPSLA 2024)

Senior Software Engineer

Strava

San Francisco, CA

August 2019 – July 2022

- * Led platform-wide migration of core activities data model (1B+ records) into a centralized microservice to unlock new sport types (trail running, MTB, etc.)
 - * Investigated malformed route data, identifying/fixing root causes and then shipping Apache Spark backfill job and request hooks to remediate affected data
 - * Used Apache Kafka and Spark to build geospatial data pipeline powering Strava Metro, a platform used by transportation agencies worldwide for urban planning
 - * Developed Scala microservice using Finatra, Apache Thrift, and Cassandra to serve millions of static map rendering requests across iOS, Android, and web
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Education

Cornell University Ann S. Bowers College of Computing and Information Science

PhD in Computer Science

New York, NY

August 2024 – Present

University of Pennsylvania School of Engineering and Applied Science

MSE in Computer and Information Science

Philadelphia, PA

August 2022 – May 2024

University of Pennsylvania School of Engineering and Applied Science

BSE with Honors in Computer and Information Science

Philadelphia, PA

August 2015 – May 2019

Skills

Programming Languages: OCaml, Scala, Python, Haskell, Java, Ruby, C/C++

Software Engineering: Git, Rails, Apache Spark, Apache Kafka, Apache Thrift, Cassandra, MySQL

Formal Verification: Agda, Rocq, Lean

Publications

Weighted NetKAT: A Programming Language for Quantitative Network Verification

PLDI 2026

Emmanuel Suárez Acevedo, Tiago Ferreira, Kevin Batz, Oliver Bøving, Nate Foster, and Alexandra Silva

Effects and Coeffects in Call-By-Push-Value

OOPSLA 2024

Cassia Torczon, Emmanuel Suárez Acevedo, Shubh Agrawal, Joey Velez-Ginorio, and Stephanie Weirich

Awards

Graduate Research Fellowship, National Science Foundation (2024)

Outstanding Research Award, University of Pennsylvania SEAS (2024)

Dean's Excellence/Hopper-Dean Fellowship, Cornell Bowers CIS (2024)
